Virtual University Overseas Examination System V2.1 (x86)

**Numerical Analysis (MTH603)**

**Question: 1 (Marks: 1)**

Maximum number of complex roots in an equation of degree 3 are

**Choices:**

- 2
- 3
- 0
- 1

Math knowledge

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 2 (Marks: 1)

Complex roots of a quadratic equation are ----------of each other.

Choices:

- Equal
- Negative
- Reciprocal
- Conjugate

Math knowledge

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60

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 3 (Marks: 1)

If S is a symmetric matrix then

Choices:

- $S^{-1} = S^T$
- $S^{-1} = S$
- $S^T = S$
- All are true

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Mathknowledge

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59

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**Numerical Analysis (MTH603)**

**Question: 4 (Marks: 1)**

Diagonal dominance of a coefficient matrix is strictly checked in

**Choices:**

- Muller's method.
- Bisection method
- Jacobi's method
- Newton-Raphson method

Math knowledge

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Virtual University Overseas Examination System V2.1 (x86)

**Numerical Analysis (MTH603)**

**Question: 5 (Marks: 1)**

Coefficient matrix[A] of system of equations can be decomposed into product of [L] and [U] in

**Choices:**

Guassian elimination method

Gausss Jordan elimination method

Crouts reduction method

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All methods mentioned here

Math knowledge

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57

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Numerical Analysis (MTH603)

Question: 6 (Marks: 1)

Jacobi's method is highly recomended for....

Choices:

- Real and symmetric matrix.
- non-symmetric matrix.
- Complex symmetric matrix.
- None of the given choices

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Math knowledge

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55

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 7 (Marks: 1)

In power method, initial vector is chosen such that the largest element is

Choices:

1

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2

3

4

Math knowledge

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54

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 8 (Marks: 1)

In Central difference table, all the odd differences have a ..... Suffix.

Choices:

- Whole number
- Fractional
- Real number
- None of these

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Math knowledge

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53

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 9 (Marks: 1)

For a given table of values, the process of estimating the value of y, for any intermediate value of x , is known as

Choices:

- Interpolation
- Extrapolation
- Forward difference
- Backward difference

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Math knowledge

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52

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Numerical Analysis (MTH603)

Question: 10 (Marks: 1)

Forward difference operator is defined for

Choices:

equally spaced abscissas .

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unequally spaced abscissas.

constant abscissas.

None of the above .

TIME LEFT
51

Math knowledge

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Numerical Analysis (MTH603)

Question: 11 (Marks: 1)

All odd central differences have ---------suffix

Choices:

- Prime
- Even
- Odd
- Fractional

Math knowledge

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50

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Numerical Analysis (MTH603)

Question: 12 (Marks: 1)

When shift operator E operates on f(x), the result is the --------value of the function.

Choices:

- first
- previous
- next
- last

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Math knowledge

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49

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Numerical Analysis (MTH603)

Question: 13 (Marks: 1)

In Muller's method $f(x) = 0$ is approximated by a

Choices:

- First degree polynomial
- Second degree polynomial
- Third degree polynomial
- Fourth degree polynomial

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Math knowledge

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47

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**Numerical Analysis (MTH603)**

**Question: 14 (Marks: 1)**

Hexadecimal describes a base ---------- number system.

**Choices:**

16

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8

10

2

TIME LEFT
46

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Math knowledge

Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 15 (Marks: 1)

The convergence of which of the following method is sensitive to starting value?

Choices:

- False position
- Gauss Seidel method
- Newton-Raphson method
- Bisection Method

Math knowledge
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45

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**Numerical Analysis (MTH603)**

**Question: 16 (Marks: 1)**

Secant Method is also called as?

**Choices:**

- 2-point method

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- 3-point method
- 4-point method
- 5-point method

TIME LEFT

Math knowledge

Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 17 (Marks: 1)

Newton's backward difference formula is used for interpolating the values of y _____

.

Choices:

a short distance backward from $y_0$

a short distance forward from $y_n$

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near the end of table of values

near the beginning of a set of values

Math knowledge

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43

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Numerical Analysis (MTH603)

Question: 18 (Marks: 1)

P in Newton's backward difference formula is defined as _____ .

Choices:

- $P = \left(\frac{x+x_0}{h}\right)$
- $P = \left(\frac{x-x_n}{h}\right)$
- $P = \left(\frac{x-x_0}{h}\right)$
- $P = \left(\frac{x+x_n}{h}\right)$

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Math knowledge

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41

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**Numerical Analysis (MTH603)**

**Question: 19 (Marks: 1)**

Newton's forward difference formula is used for extrapolating the values of y _____

**Choices:**

- A short distance forward from $y_n$
- Near the end of set of values
- Near the beginning of a set of values
- A short distance backward from $y_0$

Math knowledge

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39

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 20 (Marks: 1)

P in Newton's forward difference formula is defined as _____ .

Choices:

- $P = (\frac{x-x_0}{h})$
- $P = (\frac{x+x_n}{h})$
- $P = (\frac{x-x_n}{h})$
- $P = (\frac{x+x_0}{h})$

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Math knowledge

TIME LEFT

38

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**Numerical Analysis (MTH603)**

**Question: 21 (Marks: 1)**

In Relaxation method, if no starting solution is given, then the starting solution should be taken as _____ .

**Choices:**

- (1, 0, 0)
- (0, 1, 0)
- (1, 1, 1)
- (0, 0,0 )

Math knowledge

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35

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Numerical Analysis (MTH603)

Question: 22 (Marks: 1)

Gauss-Seidal iterative method is used to find solution of _____ .

Choices:

- Non linear inequality
- Non linear equation
- Linear Equation
- Linear inequality

Math knowledge

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34

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Numerical Analysis (MTH603)

Question: 23 (Marks: 1)

The quantity of error present in the statement of the problem itself, before finding its solution is called

Choices:

- Typing error
- Local round off error
- Inherent error
- Local Truncation error

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Math knowledge

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30

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Numerical Analysis (MTH603)

Question: 24 (Marks: 1)

The Newton - Raphson method formula for finding the square root of a positive number $N$ from the equation $x^2 - N = 0$ is .....

Choices:

$x_{i+1} = \frac{1}{2}\left(x_i + \frac{N}{x_i}\right)$

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$x_{i+1} = \frac{3x_i}{2}$

$x_{i+1} = \frac{1}{2}\left(3x_i - \frac{N}{x_i}\right)$

$x_{i+1} = \frac{x_i}{2}$

Math knowledge

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28

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 25 (Marks: 1)

Every square ------- matrix has an inverse.

Choices:

- Zero
- Singular
- None of the above
- Non-singular

Math knowledge

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26

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Numerical Analysis (MTH603)

Question: 26 (Marks: 1)

When $\lambda$ is the eigen value and $X$ is the corresponding eigen vector then_____ .

Choices:

$[A]^{-1}X = \lambda X$

$[A]X = \lambda X$

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$[A]^2X = \lambda X$

$[A]X = 2\lambda X$

Math knowledge

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25

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Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 27 (Marks: 1)

The relationship between central difference operator and the shift operator is given by _____ .

Choices:

$\delta = E + E^{-1}$

$\delta = E^{\frac{1}{2}} + E^{-\frac{1}{2}}$

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$\delta = E - E^{-1}$

None of these.

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25

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Math knowledge

Virtual University Overseas Examination System V2.1 (x86)

**Numerical Analysis (MTH603)**

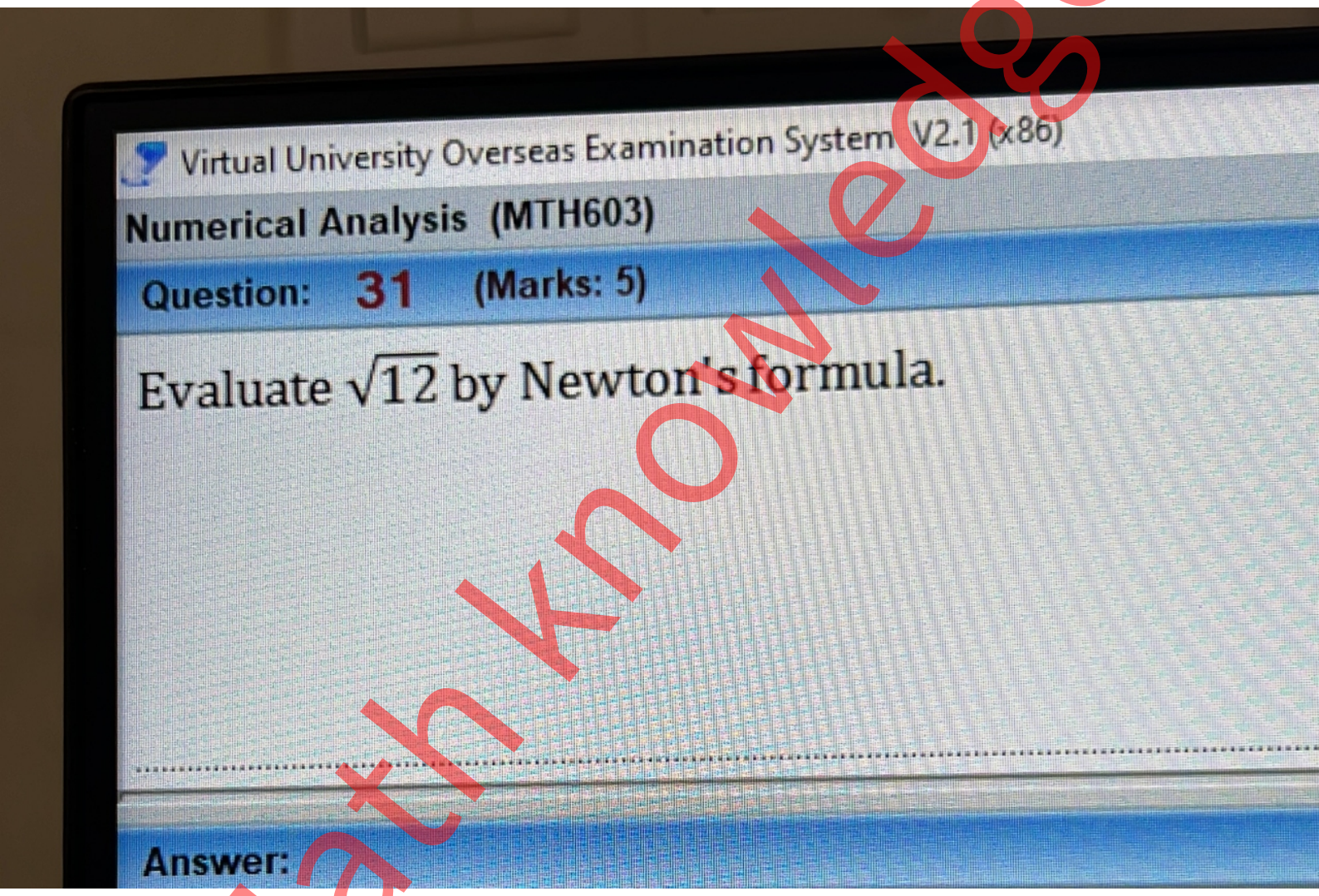
**Question: 30 (Marks: 5)**

Solve the following system of equations by Gauss-Seidal iterative method to four decimal places and up to one iteration

$$\begin{aligned} 8x+y-z &= 8 \\ x-7y+2z &= -4 \\ 2x+y+9z &= 12 \end{aligned}$$

Taking the initial solution as $(0,0,0)^T$

Math Knowledge




Math knowledge

Virtual University Overseas Examination System V2.1 (x86)

Numerical Analysis (MTH603)

Question: 28 (Marks: 3)

Define Shift operator.

Answer:

Shift operator is an operator that takes a function to its translation.
Shift operator are example of linear operators which is important for simplicity and natural occurance.

Math knowledge

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8

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Answer:

$$x = \frac{1}{8}[z - y + 8]$$

$$y = \frac{1}{7}[x + 2z + 4]$$

$$z = \frac{1}{9}[-2x - y + 12]$$

*first iteration*

$$x = \frac{1}{8}[0 - 0 + 8] = \frac{8}{8} = 1$$

$$y = \frac{1}{7}[0 + 0 + 4] = \frac{4}{7}$$

$$z = \frac{1}{9}[-0 - 0 + 12] = \frac{12}{9}$$

*second iteration*

$$x = \frac{1}{8}[1.33 - 0.5714 + 8] = 1.2377$$

$$y = \frac{1}{7}[1 + 2(1.33) + 4] = 1.0942$$

$$z = \frac{1}{9}[-2(1) - 0.5714 + 12] = 1.7488$$

Math knowledge

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5

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